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Original research article

Omni-junction silicon modulators of unequal proportions for high-speed datacom

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ABSTRACT

Doping profiles are always a key factor in the design of silicon modulators. As the doping in the waveguide cross-section is limited by the compact footprint, three-dimensional complex optimization presents itself as a viable option for improved performance. This work proposes two different types of unequal-proportioned omni-junction modulators obtained by the linear-decline weight particle swarm optimization algorithm with appropriate figure of merits (FOM). With FOM targeting at overall performance, the obtained modulator composed of unequal-proportioned vertical and lateral junctions has high bandwidth of 40 GHz with the $V_{\pi}L$ of 1.07 V cm. With FOM for higher modulation efficiency, the modulator composed of unequal-proportioned U-shaped and L-shaped junctions demonstrates low $V_{\pi}L$ of 0.72 V cm and bandwidth maintaining over 36 GHz. The algorithm together with the 3D design method provides greater flexibility in improving the device performance.

1. Introduction

Emergence of high-density silicon photonics (SiP) for optical communications is revolutionizing data transmission over short and long distances, enabling unprecedented levels of efficiency, speed and cost-effectiveness. Due to strong light confinement, SiP has advantages of complementary metal-oxide-semiconductor (CMOS) compatibility and much more compact footprint, which makes SiP an attractive option for high-speed applications [1]. Modulators are the core of all the components in the optical transmission link, enabling the conversion of electrical signals into light signals, which can then be transmitted over long distances without significant loss or interference [2,3]. Capability of operating at high speeds and low power consumption makes silicon modulators suitable for using in telecommunications networks [4–6]. Silicon modulators are generally based on the plasma dispersion effect, among which Mach–Zehnder modulators (MZMs) provide superior thermal stability and better fabrication tolerance [7] than microring modulators (MRMs), making MZMs the preferred choice in practical commercial applications. Since the optical modulation in SiP is accomplished by effectively manipulating the charge density in the rib waveguide via driving voltages, the doping profile plays an important role. Various types of doping profiles have been studied especially for the carrier-depletion modulator, such as lateral junctions [8], vertical junctions [9], L-shaped junctions [10] and so on [11]. Although doping profiles can be adjusted in the rib cross-section, this is limited by the size of the rib waveguide. In order to improve modulation efficiency, it is proved to be an effective

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method to generate junctions in the optical propagation direction [12–18]. Therefore, the performance of the modulator can be significantly improved by optimizing the doping profile in three dimensions while generating p–n junctions in the cross-section and propagation direction.

In this paper, we propose two types of omni-junction silicon modulators with the linear-decline weight particle swarm algorithm for better optimization. The innovative design of the omni-junction modulators is the first proposed composition of unequal-proportioned doping profiles in the propagation direction. Thus this modulator allows for the optical signal to be modulated in multiple directions, resulting in improved signal transmission and enhanced performance. The modulator composed of the unequal-proportioned vertical and lateral junctions has the $V_\pi L$ of 1.07 V cm and the estimated bandwidth of 40 GHz with the loss being 16.6 dB/cm. The modulator composed of the unequal-proportioned U-shaped and L-shaped junctions has the $V_\pi L$ of 0.72 V cm and the estimated bandwidth of 36 GHz with the loss being 18.7 dB/cm. Additionally, the omni-junction modulators have low power consumption and can be manufactured using standard fab processes. These features make the designed modulators attractive choices for optical communication systems.

2. Methods

To achieve precise and accurate results, the Monte-Carlo method is significant for simulating implantation [19–21]. Thus we utilize the effective 3D Monte-Carlo method which is put forward in our previous work [22] to generate complex 3D doping profiles in the rib waveguide. The effective 3D Monte-Carlo method is fulfilled by obtaining and mapping 2D Monte-Carlo doping profiles in three dimensions to save computation time and resources while maintaining the accuracy of the whole doping profile. This approach facilitates the production of high-performance omni-junction modulators, which are defined as the modulators with complex doping profiles with p–n junctions generated in three dimensions. Fig. 1 shows the structure of the 3D modulator to be optimized. Fig. 1(a) is the top view of the 3D-optimized MZM whose modulating arm consists of multiple doping periods. The region between the black dashed lines is a period of length L_p, which is set to be 1.2 μm in this work. By simulating the halved length for each doping in a period, the same performance can be achieved which saves time and resources. Thus the simulation can be greatly simplified due to its periodic structure. The periodic structure for simulation is denoted by Fig. 1(b), which has two doping parts, named the former doping and the latter doping, respectively. The doping for the former part has an active region length of L_{pd} and the latter doping region length is $L_p - L_{pd}$. The cross-section of the 3D MZM is shown in Fig. 1(c), based on the typical 220-nm SOI wafer whose buried oxide (BOX) is 2 μm. The width of the rib waveguide is 410 nm and the height of the slab is 60 nm to satisfy the single-mode condition. The tilt angle of the implantation for simulation is 0° and the rapid thermal annealing is at 1030 °C within 10 s. The parameters of the desired complex 3D doping profile to optimize are doping doses, doping energies and doping locations. To accommodate to the practical application, the doping parameters are optimized within the limits of the ordinary two-step manufacturing process. The doping doses are within the span of 10^{12}∼10^{14} which is appropriate for low-loss and high-efficiency design. The doping energies are within the span of 0∼70 keV for P-type doping and within the span of 0∼140 keV for N-type doping as ordinary fab process required. The doping positions are optimized both in the propagation direction and in the cross-section, namely the L_{pd} in Fig. 1(b), and the X_N and X_P in Fig. 1(c). As in interleaved modulators based on pure P-type and pure N-type doping, different doping types account for the same length since the depletion region expansion mainly takes place in the Z direction, and thus the unequal proportion of P-type and N-type doping reduces efficiency. While for the omni-junction modulator, the proportion of parts in a period has a great influence on the whole performance as each part itself modulates light. The overall performance of the omni-junction is determined by the composition of two parts.

To optimize the doping parameters systematically for the purpose of achieving excellent performance as decreasing $V_\pi L \times Loss$ and broadening EO bandwidth, we adopt the particle swarm optimization (PSO) algorithm, whose flow chart is displayed in Fig. 2. The PSO algorithm is capable of finding optimal results defined on a multidimensional vector space with fast convergence, making it suitable for optimizing tens of doping parameters in this work. With the PSO algorithm applied, the number of variable doping parameters is the number of dimensions in the vector space and each doping parameter acts as a dimension of the vector position of the particle. Together the particle has the vector position as x. The results are compared with regard to the figure of merit (FOM), which is the fitness (F) of the PSO algorithm and is defined as

$$F(x) = FOM = \theta V_\pi L + \frac{V_\pi L \cdot Loss}{RC^2} \tag{1}$$

$Loss$ is the insertion loss and we mainly consider the carrier-induced loss in this work, which is much higher than the propagation loss in the silicon waveguide. R and C are the resistance and capacitance of the 3D junction, which are obtained via optoelectronic simulation by TCAD tools. From RC parameter, the EO bandwidth can be estimated. Thus the FOM has $V_\pi L$, Loss and EO bandwidth taken into consideration. To obtain the modulator with high modulation efficiency and high bandwidth with acceptable loss, we have $V_\pi L$ emphasized as θ.

With the fitness, at the iteration i, the local optimal result $P_{loc}(i)$, namely the optimal result in the current iteration, is defined as

$$P_{loc}(i) = min(F(x(i))) \tag{2}$$

The global optimal result $P_{glo}(i)$, namely the optimal result in all iterations that have been completed so far, is defined as

$$P_{glo}(i) = min(F(x)) \tag{3}$$

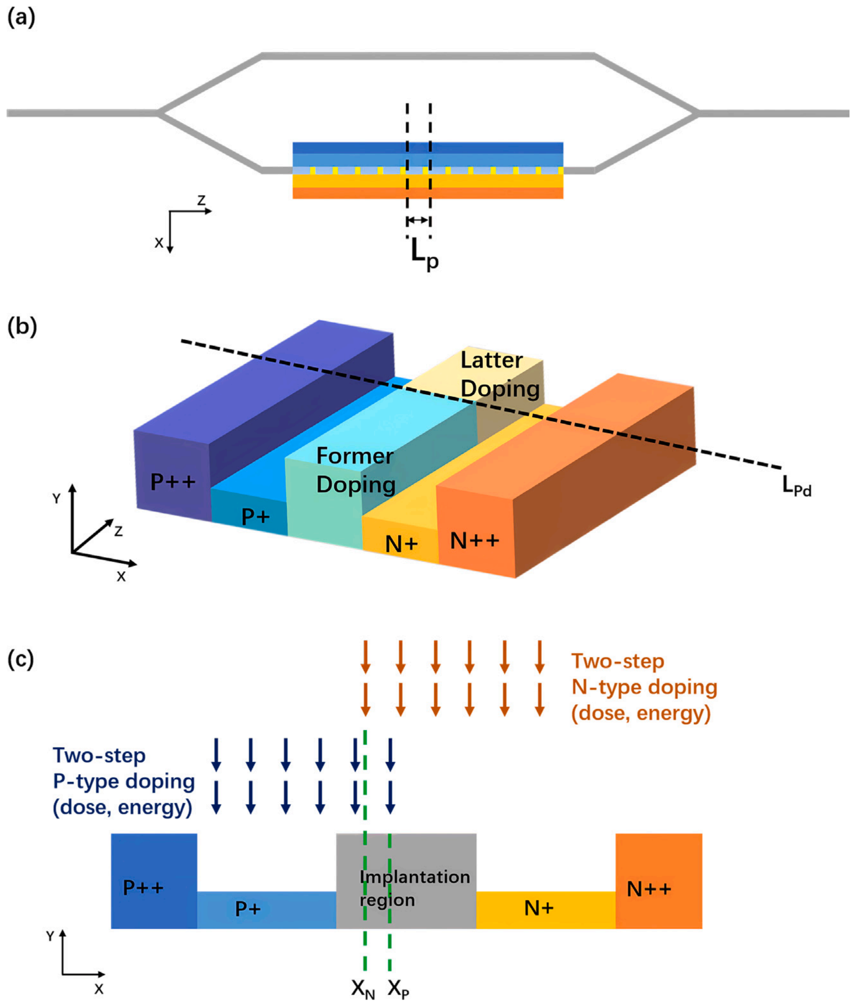

Fig. 1. (a) The top view of the 3D MZM consisting of periods whose lengths are all L_p. (b) single omni-junction period with unequal-proportioned doping. (c) the cross-section of the modulator with two-step doping process, X_N and X_P are the doping locations in the X direction of N-type and P-type doping, respectively.

By comparing the fitness of each particle at the current iteration, the positions of particles are updated according to the optimizing velocity vector v [23], which is

$$v(i+1) = \omega_m \cdot v(i) + c_{loc} \cdot r_1 \cdot (P_{loc}(i) - F(x(i))) + c_{glo} \cdot r_2 \cdot (P_{glo}(i) - F(x(i))) \tag{4}$$

where $x(i)$ is the particle position at the current iteration i and the updated position becomes

$$x(i+1) = x(i) + v(i+1) \tag{5}$$

In the velocity vector, c_{loc} and c_{glo} are constants as the personal and social influence of the local and global best positions, which both equal 2 in this work. r_1 and r_2 are two random values in the span of [0,1]. ω_m is the inertia weight. The performance of the standard PSO algorithm depends largely on the selection of the inertia weight. To avoid the confinement that standard PSO algorithm tends to generate oscillations near the global optimization solution [24], especially for the application scene in this work that the minor variation in the doping profiles possibly causes large performance change, the linear-decline inertial weight ω_m is adopted as

$$\omega_m = \omega_{max} - \frac{i}{I}(\omega_{max} - \omega_{min}) \tag{6}$$

where ω_{max} is 0.9. ω_{min} is 0.3. I is the total number of iterations to complete and we iterate 30 times for convergence. Hence, we applied the linear-decline weight particle swarm optimization (LDWPSO) [23] algorithm to accelerate convergence, thus balancing exploration speed and accuracy.

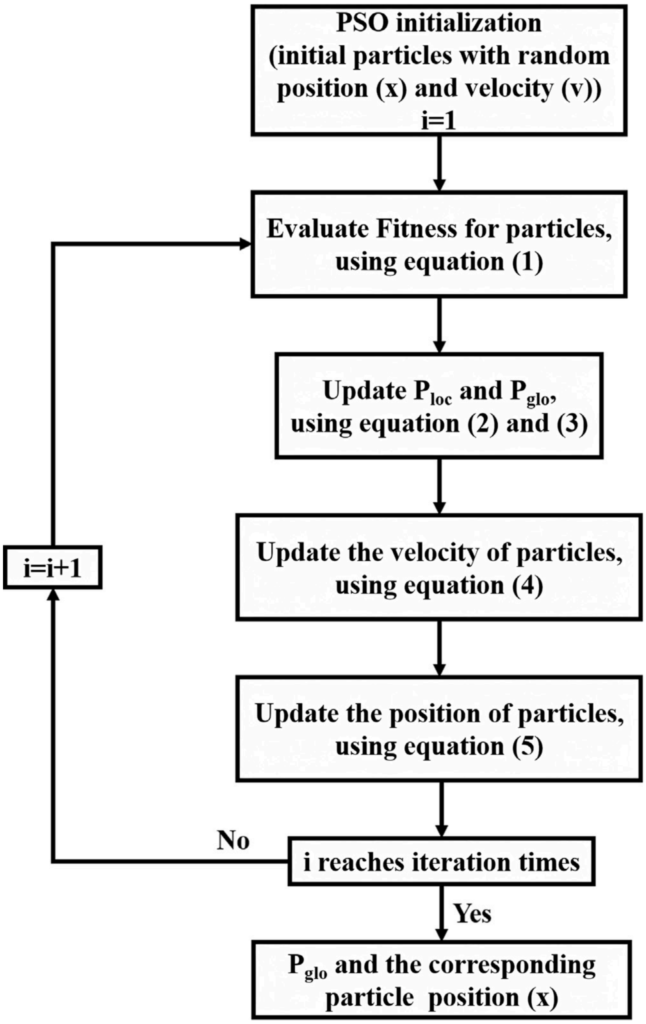

Fig. 2. The flow chart of PSO algorithm.

Table 1
Implantation parameters of the vertical junction.

Steps	Species	Dose (cm^{-2})	Energy (keV)	Window (μm)
1	Phosphorus	5e12	140	(−0.205, 3)
2	Phosphorus	5e12	10	(−0.205, 3)
3	Boron	6e12	10	(−3, 0.205)
4	Boron	5e12	35	(−3, 0.205)

3. Result and discussion

3.1. Lateral + vertical

With LDWPSO algorithm and $\theta = 10$ in FOM to have both bandwidth and modulation efficiency taken into account in almost equal measure, the optimized omni-junction silicon modulator for the C band is displayed in Fig. 3, which is composed of vertical-junction part and lateral-junction part. Their implantation parameters are shown in Tables 1 and 2. The doping windows in the tables are based on the coordinate system with the origin at the rib center.

The former doping forms the vertical junction and the latter doping forms the lateral junction. The doping level is set at around 1e18 cm^{-3} to ensure that the efficiency, loss and impedance are balanced. The optimal proportion of the vertical-junction part to the lateral-junction part in a period is 3:2. The simulation results of equivalent $V_{\pi}L$ and carrier-induced insertion loss of the designed

Table 2
Implantation parameters of the lateral junction.

Steps	Species	Dose (cm^{-2})	Energy (keV)	Window (μm)
1	Phosphorus	2e13	80	(−0.1, 3)
2	Phosphorus	2e13	70	(−0.1, 3)
3	Boron	9e12	50	(−3, 0.068)
4	Boron	2e13	35	(−3, 0.068)

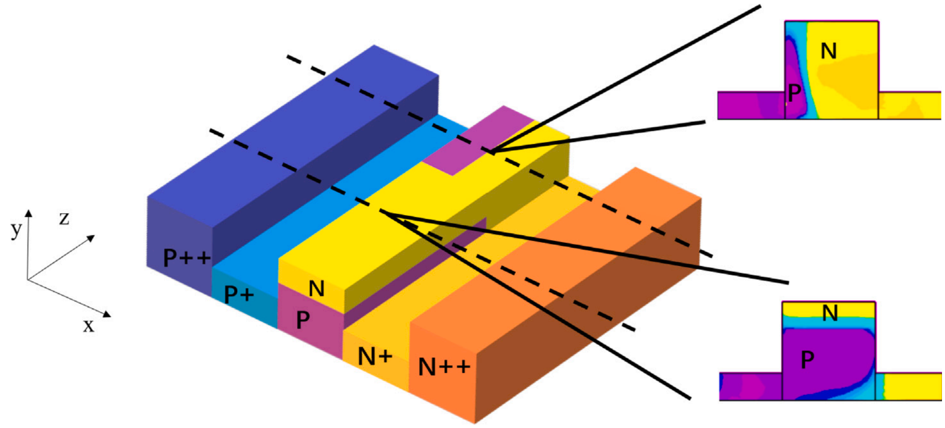

Fig. 3. Schematic diagram of the designed modulator period based on vertical and lateral junctions.

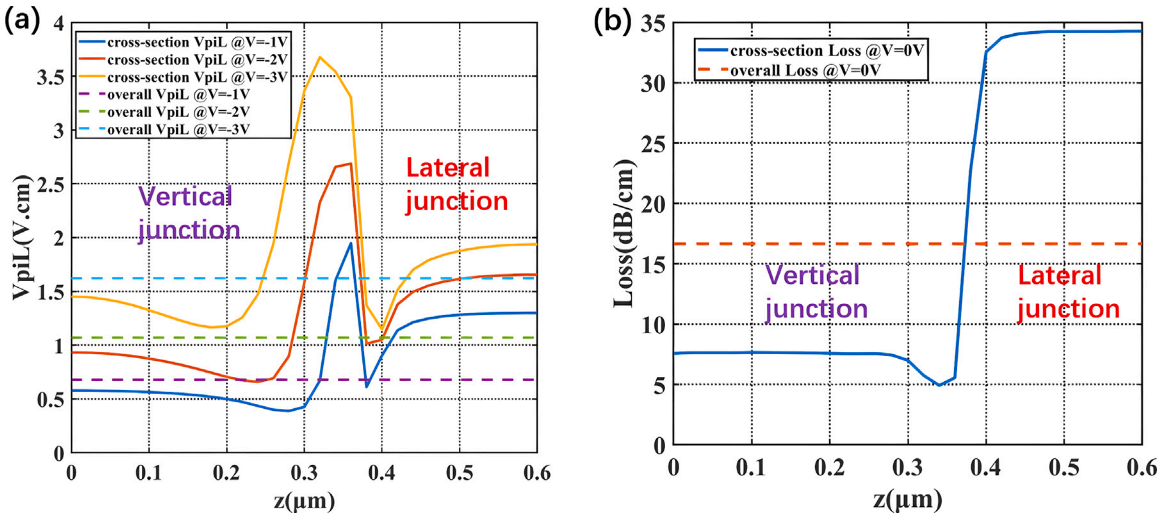

Fig. 4. (a) $V_\pi L$ and (b) carrier-induced insertion loss of the modulator based on the unequal-proportioned vertical and lateral junctions under reverse biases. Solid lines are the performance along the propagation direction (z). Dashed lines are the overall performance of the period.

modulator are shown in Fig. 4. By Vpp being 2 V, the equivalent $V_\pi L$ of the period is 1.07 V cm at the bias of −2 V and 1.6 V cm at the bias of −3 V. The carrier-induced insertion loss is 16.6 dB/cm.

The solid lines in Fig. 4 describe the variation of $V_\pi L$ and loss along the propagation direction, respectively. The vertical-junction part provides higher modulation efficiency and much lower loss than the lateral-junction part. The reason for the great performance is that the vertical junction located right above the mode field center has donor concentration twice as high as acceptor concentration. The increased doping of the donor results in a larger depletion region expansion extending across the mode center, making modulation much more efficient. The lower doping of the acceptor results in lower loss. As a result, the length of the lateral-junction part has been decreased in order to take full advantage offered by the vertical-junction part. However, the lateral-junction part is also important to ensure proper RF performance as this part supports the current flow, which must be of sufficient size to reduce resistance. The P-N overlap of these two parts in the propagation direction covers large areas, which contributes to the decline of $V_\pi L$ as demonstrated by dips of solid lines in Fig. 4(a).

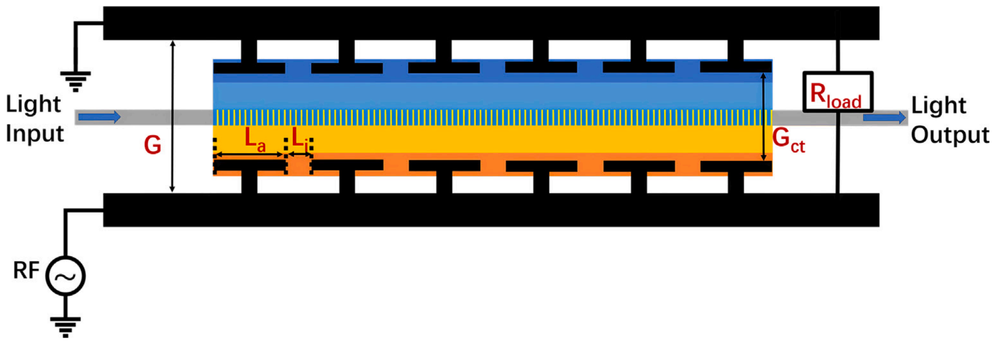

Fig. 5. The top view of the modulator arm with the segmented transmission line.

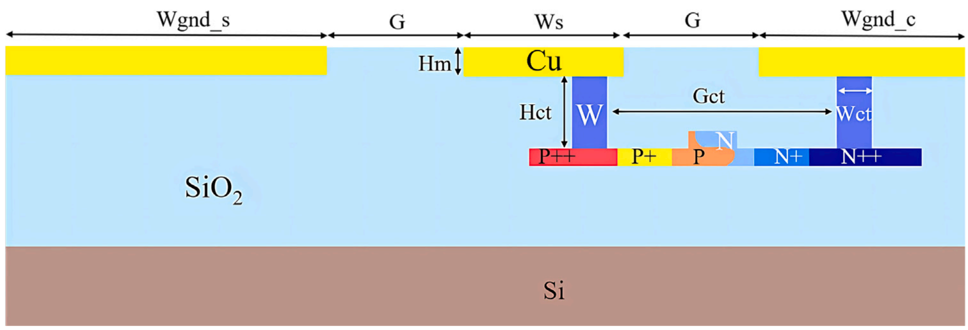

Fig. 6. The cross-section of the modulator with the segmented transmission line.

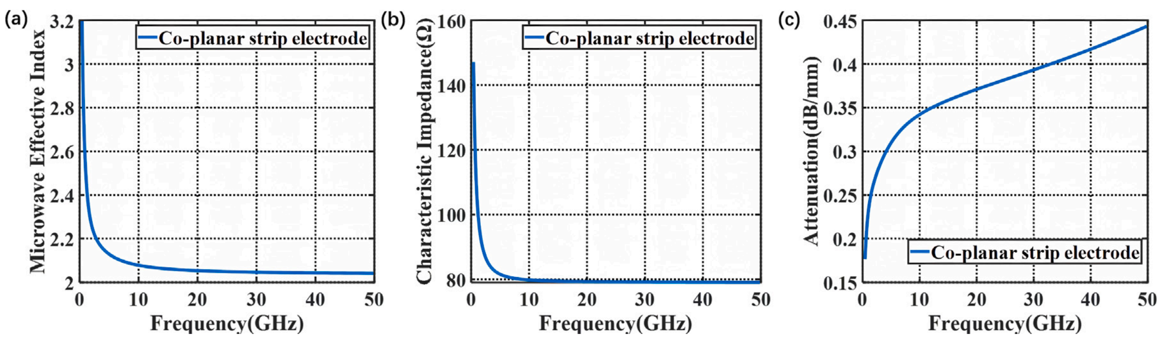

Fig. 7. (a) Microwave effective index, (b) characteristic impedance, and (c) microwave attenuation of the segmented transmission line.

The RF performance of the proposed modulator is estimated by using the segmented transmission line which can be fabricated in a CMOS foundry. The segmented transmission line is applied to reduce the microwave velocity and impedance for higher operating frequencies. The structure of the segmented transmission line is shown in Fig. 5. L_a is the modulation segment length which is 8.4 μm to match the omni-junction period and L_i is the interval between each segment electrode, being 1.6 μm. Gct is the gap between contact holes and G is the gap between the copper lines. R_{load} is the impedance of the terminator to match the impedance of the loaded transmission line. The cross-section of the transmission line is denoted by Fig. 6. The transmission line parameters are that $Wgnd\_s = 155$ μm, $Ws = 5$ μm, $Wgnd\_c = 92$ μm, $G = 8$ μm, $Hm = 0.85$ μm, $Hct = 1.1$ μm, $Gct = 8$ μm and $Wct = 3$ μm. The effective index, unloaded characteristic impedance and attenuation of the segmented transmission line are shown in Fig. 7.

The bandwidth is estimated based on the microwave transmission (ABCD) matrix theory [25,26]

$$S_{21} = \left| \frac{\Sigma V_j}{V_{pp}/2 \times \Sigma L_a \times (1 + j\omega R_{pn} C_{pn})} \right| \tag{7}$$

where V_j is the voltage on the PN junction [27], considering impedance mismatch, microwave loss, microwave dispersion effects and velocity mismatch. V_{pp} is the driving signal voltage, being 2 V. At bias of −3 V, $R_{pn} = 11$ kΩ μm and $C_{pn} = 0.25$ fF/μm. The low R_{pn} is attributed to the current path provided by the lateral-junction part in the propagation direction which transports electrons

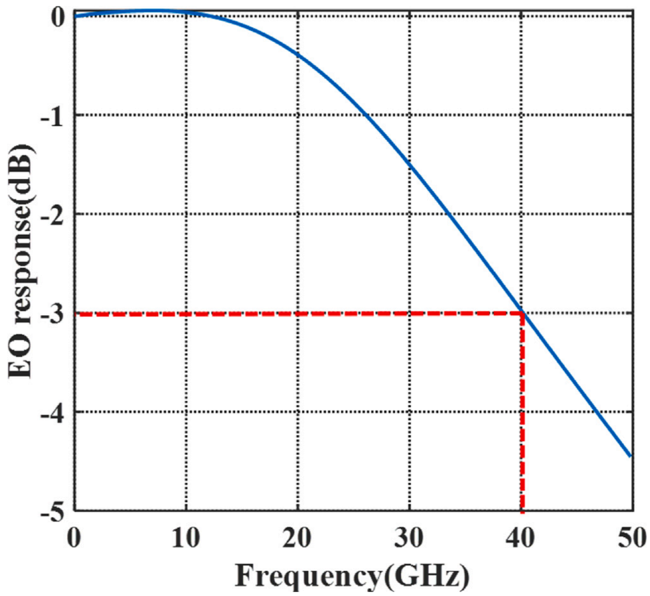

Fig. 8. The frequency response of the modulator based on the unequal-proportioned vertical and lateral junctions at −3 V bias.

Table 3
Implantation parameters of the U-shaped junction.

Steps	Species	Dose (cm^{-2})	Energy (keV)	Window (μm)
1	Phosphorus	1e13	10	(−0.13, 3)
2	Phosphorus	2e13	90	(−0.13, 3)
3	Boron	6e12	65	(−3, 0.16)
4	Boron	2e13	15	(−3, 0.16)

Table 4
Implantation parameters of the L-shaped junction.

Steps	Species	Dose (cm^{-2})	Energy (keV)	Window (μm)
1	Phosphorus	2e13	140	(−0.05, 3)
2	Phosphorus	2e13	90	(−0.05, 3)
3	Boron	1e13	35	(−3, 0.205)
4	Boron	6e12	10	(−3, 0.205)

to the N-type doped region in the vertical-junction part. The low C_{pn} is due to the moderate doping level of 1e18 cm^{-3} and the appropriate area of the PN overlap in the propagation direction without introducing excess capacitance. With the impedance of the microwave source being 50 Ω and R_{load} being 33 Ω, the bandwidth of the 1-mm modulator is obtained in Fig. 8, which demonstrates the 3 dB bandwidth over 40 GHz, catering to the high-speed requirement.

The fabrication tolerance of the modulator is analyzed in Fig. 9. In the analysis, the vertical misalignment upwards, the lateral misalignment rightwards and the misalignment forward in the propagation direction are labeled as positive, respectively. Within misalignment of ± 0.1 μm, the $V_\pi L$ is mainly affected by the vertical junction misalignment and the variation is up to 30%. The loss has a fluctuation of $12 \sim 25\%$. The bandwidth variation is within 20% except for the vertical junction misalignment. From the fab tolerance analysis, margin designs of positive vertical junction misalignment, negative lateral junction misalignment and positive misalignment in the propagation direction will ensure a more stable modulator.

3.2. U-shaped + L-shaped

With $\theta = 100$ in FOM putting more emphasis on $V_\pi L$ to obtain more efficient modulation with high bandwidth and the rib waveguide width of 450 nm, the optimized structure composing of the U-shaped junction and the L-shaped junction is obtained, and the proportion of the U-shaped junction to the L-shaped junction is 2:1, as depicted in Fig. 10. The implantation parameters are provided in Tables 3 and 4. Fig. 11 illustrates the modulation efficiency and carrier-induced insertion loss of the structure. This design exhibits the low $V_\pi L$ of 0.72 V cm at −2 V and 1.28 V cm at −3 V with the insertion loss of 18.7 dB/cm. By analyzing

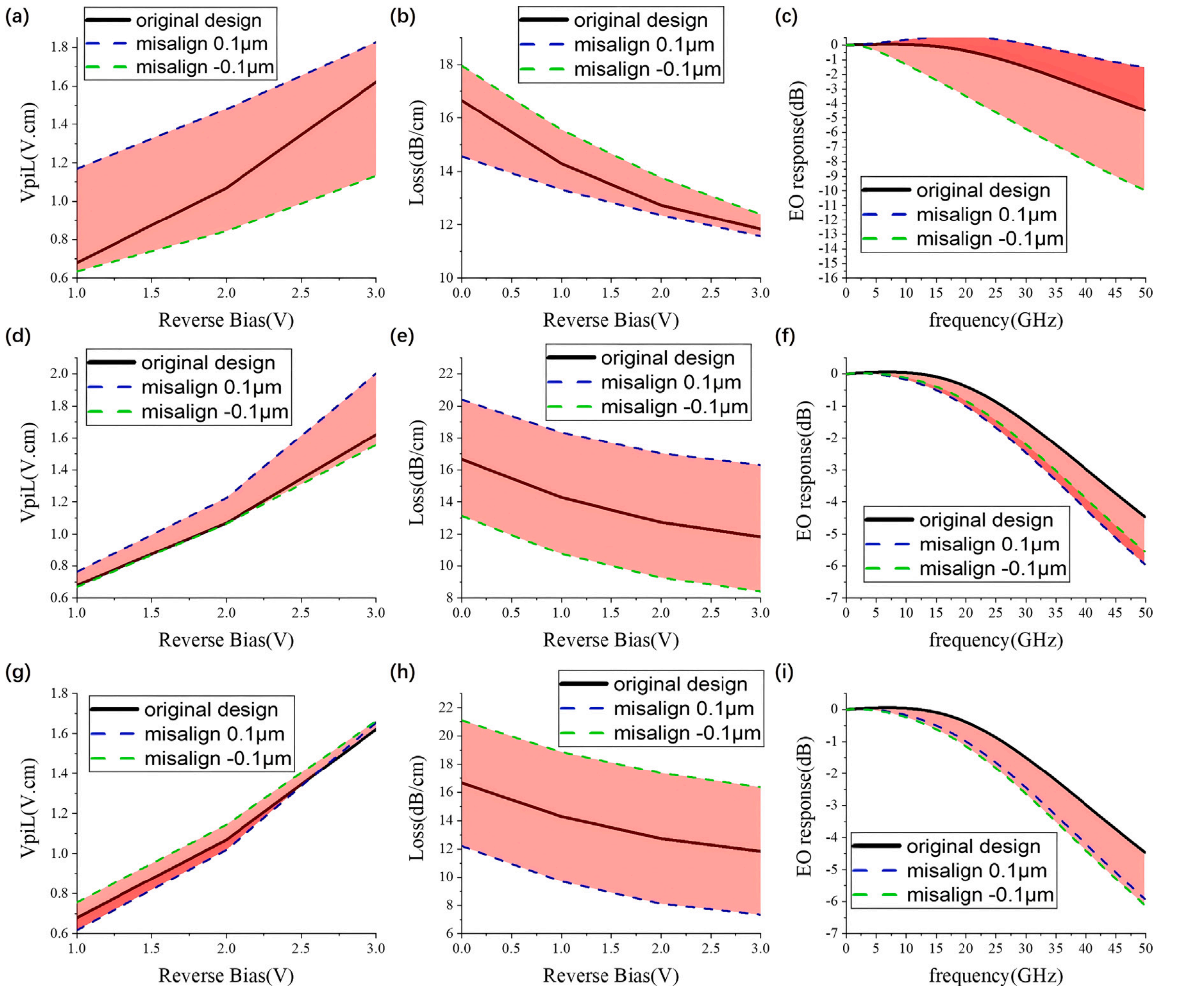

Fig. 9. The fab tolerance of the modulator based on the unequal-proportioned vertical and lateral junctions: (a) VpiL, (b) carrier-induced loss, (c) bandwidth influenced by vertical junction misalignment; (d) VpiL, (e) carrier-induced loss, (f) bandwidth influenced by lateral junction misalignment; (g) VpiL, (h) carrier-induced loss, (i) bandwidth influenced by proportional variation.

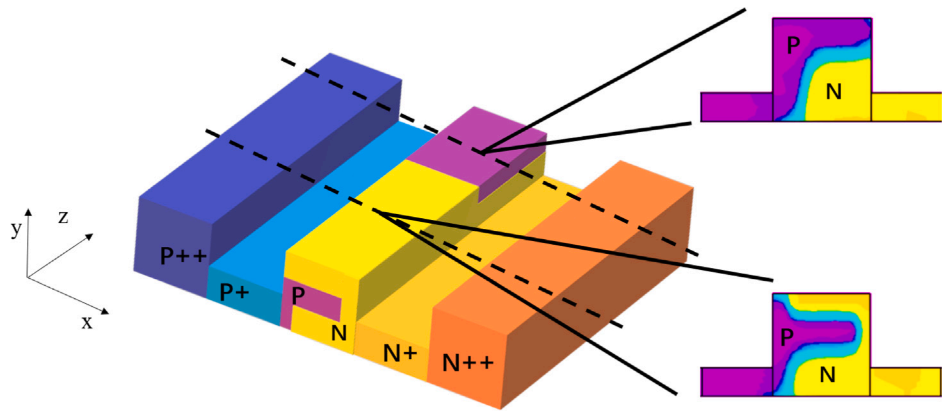

Fig. 10. Schematic diagram of the designed modulator period based on U-shaped and L-shaped junctions.

solid lines in Fig. 11, the U-shaped junction provides a much lower $V_\pi L$ which explains the reason why the U-shaped part accounts for a larger scale. Nevertheless, the current path in the cross-section of the U-shaped part is hindered by the depletion region. The

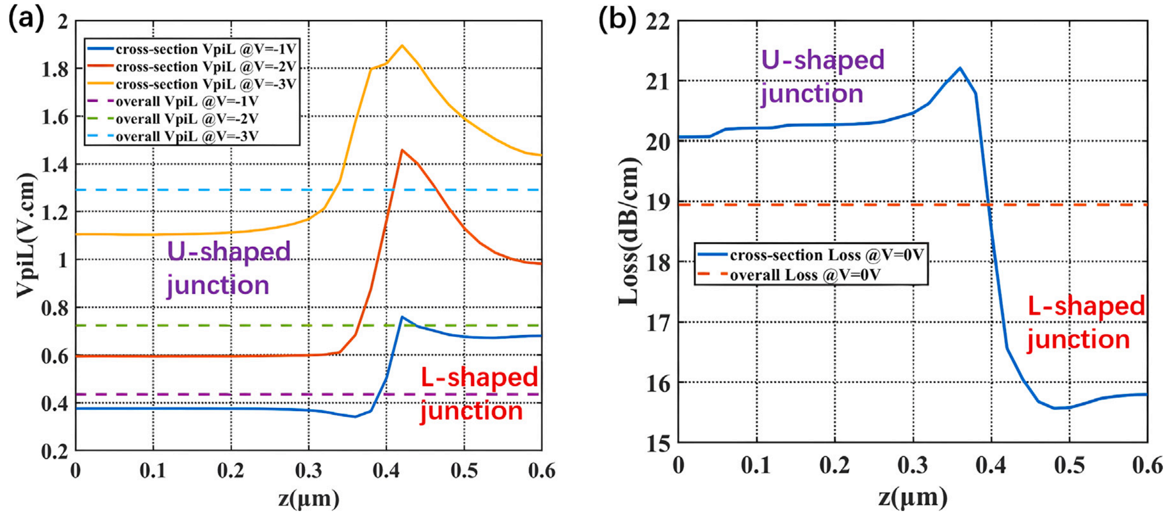

Fig. 11. (a) $V_\pi L$ and (b) carrier-induced insertion loss of the modulator based on the unequal-proportioned U-shaped and L-shaped junctions under reverse biases. Solid lines are the performance along the propagation direction (z). Dashed lines are the overall performance of the period.

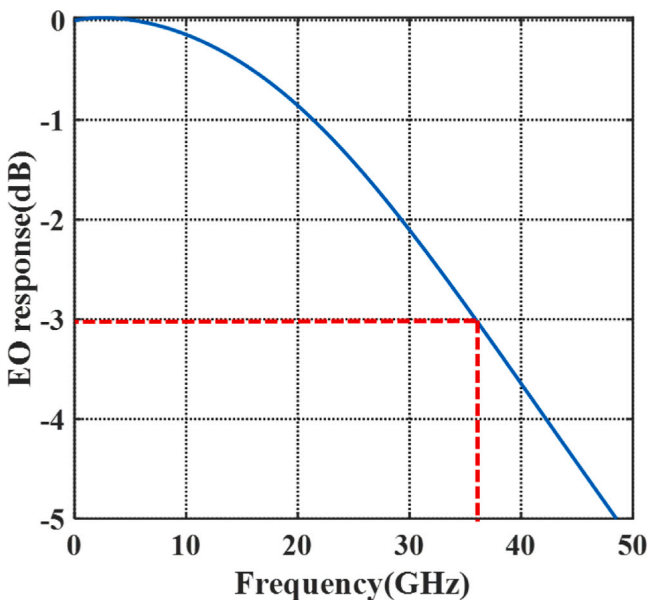

Fig. 12. The frequency response of the modulator based on the unequal-proportioned U-shaped and L-shaped junctions at −3 V bias.

L-shaped part is present to provide an alternative path for the current flow, bypassing the blocked region in the U-shaped part. Moreover, the L-shaped part also helps decrease the insertion loss of the composed period. Compared to the previous design based on vertical and lateral junctions, this design generates narrower junctions in the propagation direction and thus the declined $V_\pi L$ is mainly due to the U-shaped junction and the higher doping level. The merits of the 3D omni junction in this structure are that the superior modulation performance is supported by the U-shaped junction and the RF performance is balanced by the L-shaped junction. Since $R_{pn} = 15$ kΩ μm and $C_{pn} = 0.32$ fF/μm are obtained at bias of −3 V, the carrier transportation is well provided and the capacitance maintains at a low level despite of high carrier concentration.

With the designed segmented transmission line, the frequency response of a 700-nm modulator is estimated in Fig. 12. The 3 dB bandwidth of this design is over 36 GHz, displaying its appropriateness for datacom as well.

Fig. 13 has the fab tolerance of this design taken into consideration. The U-shaped junction misalignment upwards, the L-shaped junction misalignment upwards and the misalignment forward in the propagation direction are labeled as positive, respectively. For $V_\pi L$, the fluctuation of the U-shaped junction has manifested effect, while other misalignment causes little change. The loss variation is ±1dB/cm, which indicates that the loss is hardly affected by the misalignment. The bandwidth varies with all the

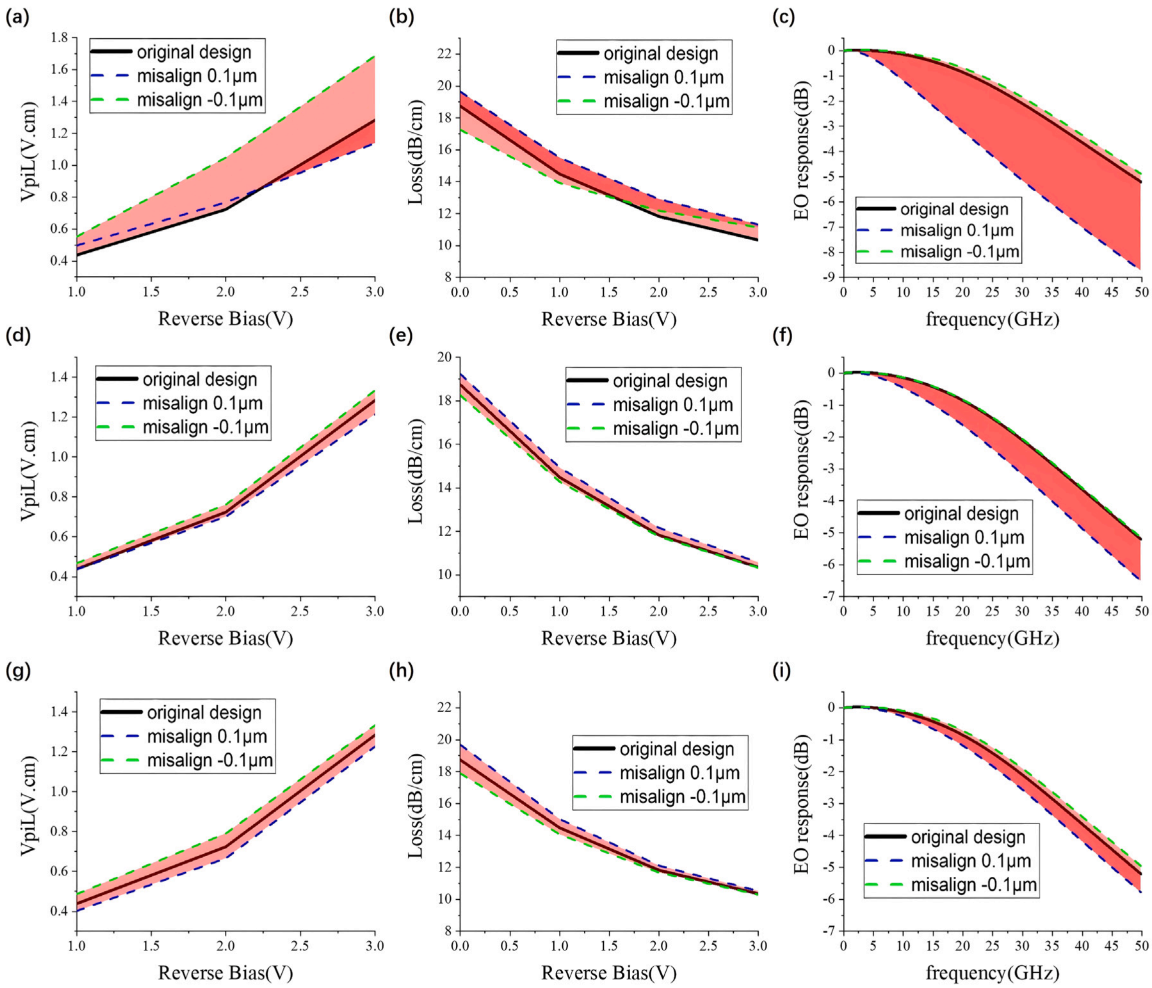

Fig. 13. The fab tolerance of the modulator based on the unequal-proportioned U-shaped and L-shaped junctions: (a) VpiL, (b) carrier-induced loss, (c) bandwidth influenced by U-shaped junction misalignment; (d) VpiL, (e) carrier-induced loss, (f) bandwidth influenced by L-shaped junction misalignment; (g) VpiL, (h) carrier-induced loss, (i) bandwidth influenced by proportional variation.

positive misalignment while the negative misalignment causes no bandwidth decline. Therefore, margin designs of negative U-shaped junction misalignment, negative L-shaped junction misalignment and negative misalignment in the propagation direction are appropriate for the modulator. The modulator composed of the unequal-proportioned U-shaped and L-shaped junctions shows better fab tolerance than the modulator composed of the unequal-proportioned vertical and lateral junctions.

Table 5 shows the comparison between our work and modulators with junction design in the propagation direction. Compared to our previous work [22], which proposes an omni-junction modulator composed of equal-proportioned vertical and lateral junctions, the improved results of our design are especially owing to the unequal proportion of two parts in a period optimized by LDWPSO, with the segmented transmission line applied. The function of each part is fully exploited by the unequal proportion of two parts. One part with a larger proportion allows for high modulation efficiency, and the other part with a smaller proportion provides the necessary current path to decrease *RC* parameters. The superior performance of these two structures in this work proves the effectiveness of 3D doping profile design that an efficient high-speed modulator can be well achieved by composing omni-junction modulators based on unequal-proportioned doping profiles. This combination has the capability of smooth and efficient signal transmission. The proposed unequal-proportioned modulators will be fabricated as our subsequent work.

4. Conclusion

To sum up, this paper proposes unequal-proportioned omni-junction silicon modulators with LDWPSO utilized for better optimization. The modulator composed of the unequal-proportioned vertical and lateral junctions has the high bandwidth of 40 GHz with the $V_\pi L$ of 1.07 V cm and loss of 16.6 dB/cm. The modulator composed of the unequal-proportioned U-shaped and L-shaped junctions has demonstrated high modulation efficiency, as the $V_\pi L$ of 0.72 V cm, and bandwidth over 36 GHz with the loss being 18.7

Table 5
Comparison with reported C-band high-speed modulators optimized in the propagation direction.

Property	Year	Result Type	$V_\pi L$ (V cm)	Loss (dB/cm)	$V_\pi L$×Loss (V dB)	Bandwidth (GHz)
[13]	2011	Experimental	2.5@-2 V	34	85	N.A. (10 Gbps)
[12]	2012	Experimental	1.7@-2 V	10	17	20@-3 V
[14]	2013	Experimental	2.4	21	50.4	20@-3 V
[17]	2018	Experimental	2.4	N.A.	N.A.	34@-3 V
[22]	2022	Numerical	0.88@-2 V	16	14.08	31.6@-3 V
Vertical+Lateral by this work	2023	Numerical	1.07@-2 V	16.6	17.76	40@-3 V
U-shaped+L-shaped by this work	2023	Numerical	0.72@-2 V	18.7	13.46	36@-3 V

dB/cm. This work reveals that via the algorithm, the 3D omni-junction design with the doping arrangement both in the propagation direction and the cross-section adds more possibility to silicon modulators. The doping structure with multiple certain-proportioned junctions has great potential for better performance, which remains as our future work.

CRediT authorship contribution statement

Zijian Zhu: Methodology, Validation, Formal analysis, Data curation, Visualization, Writing – original draft. **Yingxuan Zhao:** Software, Writing – review & editing, Funding acquisition. **Junbo Zhu:** Visualization. **Rui Huang:** Investigation. **Xiang Liu:** Validation. **Hongbao Liu:** Data curation. **Zhen Sheng:** Software, Resources. **Fuwan Gan:** Conceptualization, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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